

RAMDEOBABA UNIVERSITY



Department of Computer Science and Engineering

PROJECT SYNOPSIS

Session: _____ / *Semester:* _____ / *Programme:* _____

AI-Augmented Smart Hospital Ecosystem for Clinical Intelligence, Resource Optimization and Patient Care Management

A Queue-First, Cloud-Agnostic Platform for Multi-Hospital Operations

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1. Background and Motivation

Conventional hospital management systems are administrative record keepers. They store appointments, bills and prescriptions, but they do not manage the part of the visit a patient actually experiences, which is waiting. A patient in an outpatient department cannot reliably learn how long the wait will be, whether the doctor is present at all, or when their turn is approaching, and reception ends up answering these questions by hand, all day.

The clinical side mirrors this gap. A doctor meets a patient with no consolidated view of allergies, chronic conditions, current medications or recent results, and reconstructs that context verbally inside a consultation averaging under ten minutes. Existing products address queueing, clinical documentation and billing in isolation. A unified, cloud-agnostic platform is needed to bring these together without locking patient data into a single vendor.

2. Problem Statement

Queue state is invisible to everyone who depends on it, so waiting rooms are managed by calling out a token number rather than by any shared source of truth. Patient context is scattered across visits and departments and is assembled from memory at the moment it matters most, during the consultation itself. At the same time, existing products are either on-premise systems that cannot scale or cloud services that lock patient data into one vendor, which many institutions cannot accept for policy or regulatory reasons. A single platform is needed that keeps the queue synchronised in real time, gives the doctor a consolidated patient view before the consultation begins, and runs on infrastructure the hospital controls.

3. Objectives

- To build a real-time queue keeping the patient's phone, the doctor's workstation and the reception desk on the same state within two seconds of any change.
- To generate a consolidated one-page *patient sheet* automatically and deliver it to the doctor before the patient enters the room.
- To integrate consultation, prescription, laboratory, billing and pharmacy into one auditable workflow, removing parallel paper records.
- To apply AI as a controlled assistant for documentation, report explanation and analytics, under a strict rule that no AI output becomes a clinical record without a clinician's signature.
- To enforce privacy by design through role-based access control, per-patient ownership checks and complete audit logging.
- To deliver a cloud-agnostic system running unmodified on a managed cloud, a hospital's own Kubernetes cluster or a single server, so patient data never has to leave the institution.

4. Literature Review

Work / Source	Focus	Key Contribution and Limitation
Bailey (1952) [1]	Queueing and appointment systems in outpatient departments	Established that patient waiting time and doctor idle time are jointly optimisable through appointment scheduling policy. Purely analytical and offline; no mechanism for propagating live queue state to patients or staff.
Wolfe et al. (2006), OpenMRS [2]	Open-source modular electronic medical record	Demonstrated a deployable, extensible EMR usable across low-resource institutions. Record-keeping oriented; no real-time queue subsystem or native multi-tenancy across hospitals.
Bahga and Madiseti (2013) [3]	Cloud-based interoperable EHR	Showed that cloud-hosted EHR with standardised exchange is technically and economically viable. Assumes a single managed cloud provider; data residency and vendor lock-in remain unresolved.
Bender and Sartipi (2013), HL7 FHIR [4]	RESTful healthcare interoperability standard	Defined resource-oriented, agile exchange of clinical data between systems. Standardises the exchange format only, not queue, billing or dispensing orchestration.
Van Buchem et al. (2021) [5]; Tierney et al. (2024) [6]	Digital and ambient AI scribes	Evidence that automated documentation measurably reduces clinician documentation burden. Evaluated as an isolated tool, not integrated with queueing, prescribing or billing.
Singhal et al. (2023), Med-PaLM [7]	Clinical knowledge in large language models	Showed LLMs can reach high accuracy on clinical question answering benchmarks. Generative output is not grounded in the individual patient's record, so hallucination remains a safety risk.
Lewis et al. (2020) [8]	Retrieval-Augmented Generation	Grounding generated output in retrieved source documents materially reduces fabrication. Generic retrieval, with no notion of per-record authorization at the tool level.
Chong and Carraro (2006) [9]; Garcia-Molina and Salem (1987) [10]	Multi-tenant data architecture; Saga pattern for long-lived transactions	Defined tenancy isolation models and compensating-transaction consistency across service boundaries. Domain-agnostic; do not address global patient identity against tenant-scoped clinical records.

5. Research Gap

Identified Gap	How This Project Addresses It
Queue state exists only at the reception desk and is not shared with the patient or the clinician [1,2]	Real-time queue synchronised across patient phone, doctor workstation and reception within two seconds via WebSocket fanout with a polling fallback
Patient context is reassembled verbally at consultation time instead of being pre-delivered [2,4]	Deterministic consolidated patient sheet generated and delivered to the doctor before the patient enters the room
Queueing, clinical documentation, billing and pharmacy are solved as separate products [5,6,4]	Eight bounded-context services composing one auditable visit loop from check-in to signed prescription to dispensing
Cloud EHR designs assume a single provider, which many institutions cannot accept for data-residency reasons [3]	Cloud-agnostic deployment: one image, one Helm chart, running unmodified on managed cloud, hospital Kubernetes or a single server
AI clinical output is not reliably grounded in the patient's own record, and accountability for it is unclear [7,8]	Clinical facts retrieved at request time through typed, authorization-enforcing tools; no AI output enters the record without a clinician's signature, with full model and tool-call audit
Multi-tenancy is retrofitted rather than designed in, and cross-service consistency is left informal [9,10]	Global patient identity with hospital-scoped visit records from the first commit; database-per-service isolation with idempotent consumers and dead-letter queues

6. Proposed Methodology

The system is a domain-driven microservice platform. Thirteen bounded contexts are consolidated into eight independently deployable services, because the latency-critical path, from queue to patient sheet to notification, is the one path users watch, and every extra service boundary on it adds a network round trip and a failure mode.

Synchronous HTTP is used only where the caller needs a result to respond, such as a slot availability check. Everything else is an asynchronous domain event over a message broker with per-consumer dead-letter queues and idempotent consumers, since at-least-once redelivery is treated as normal operation rather than an error. Live queue updates reach clients over WebSocket, fanned out through a shared cache so multiple gateway replicas stay consistent, with a polling fallback for unreliable hospital networks.

Each service owns a dedicated database schema and reads no other, giving database-per-service isolation. Patient identity is global while every visit-scoped record carries a hospital identifier, producing true multi-tenancy from the first commit. Clinical facts are never generated by AI; they are retrieved from the owning service at request time through typed, authorization-enforcing tools, and the deterministic patient sheet ships first as the measured baseline any AI-generated sheet must beat before it may replace it.

Service	Responsibility
gateway	Sole public entry point: JWT verification, identity-header stripping, rate limiting, WebSocket fanout
identity	Users, credentials, roles, sessions, devices, OTP, token signing
directory	Hospitals, departments, doctors, schedules, attendance, leave, computed availability
scheduling	Appointments, waitlists, queue tokens, priority ordering, consultation state
clinical	Patient records, consultation content, prescriptions, laboratory orders and results, consent
commerce	Billing, invoices, payments, refunds, pharmacy catalogue, inventory, dispensing
comms	In-app, push, SMS, email and WhatsApp delivery behind one notification interface
ai	Agents, per-user memory, typed tool execution, evaluation harness

7. Technology, Tools and Platforms

- **Backend** – Node.js, Fastify, zod, Prisma for validation, serialization and typing from one declaration
- **Data** – PostgreSQL, Redis and a vector extension for the transactional domain and AI retrieval, without a second datastore
- **Messaging** – A message broker for routing, retries, dead-letter queues and delayed delivery
- **Object storage** – An S3-compatible store with presigned URLs, portable to any S3 provider
- **Web** – Next.js, Tailwind CSS and TanStack Query serving patient, doctor and admin route groups from one application
- **Mobile** – React Native with Expo, for over-the-air updates without a store review cycle
- **Desktop** – Electron, for printer and scanner access on reception and pharmacy machines
- **AI layer** – An orchestration framework against an OpenAI-compatible endpoint, provider-independent and able to run against a self-hosted model
- **Payments** – A payment gateway behind a provider interface, replaceable by configuration
- **Infrastructure** – Docker, Kubernetes, Helm and Terraform for one image, one chart and reproducible provisioning
- **Observability** – A self-hosted metrics, logging and tracing stack, so a customer deployment keeps full visibility

8. Expected Outcomes, Deliverables and Functional Specifications

The system delivers a patient application, covering OTP registration, discovery with live availability, appointment booking, real-time queue tracking with position and estimated wait, medical records with patient-controlled consent, and prescription and invoice downloads. The doctor workstation provides attendance, priority-ordered queue control, the automatically delivered patient sheet, structured consultation notes, and prescribing with allergy and interaction alerts. Reception and operations get walk-in registration, thermal token printing, queue control with a full audit trail, and a billing and dispensing counter that keeps working through a network outage by queueing writes locally.

The administration console covers hospital, department and fee configuration, staff and role management, and a break-glass workflow that makes clinical content reachable only with a stated reason and a distinct audit record. AI and platform services include a patient sheet agent, a clinical scribe, a triage assistant and a patient copilot, each gated by a human sign-off and an audit trail covering the model, the tool calls and whether the output was accepted or edited. A fully functional prototype demonstrating the complete visit loop, from check-in to signed prescription to invoice, is provided at the end of the project.

9. Project Scope

The system targets multi-speciality hospitals and clinic chains running outpatient departments, where queue management and clinical documentation are the binding operational constraints. It is deliberately not an inpatient system, a telemedicine platform or an insurance claims processor, and all clinical decisions remain under the control of certified professionals; AI output requires a clinician's signature before it becomes part of the record. The prototype is developed and evaluated using controlled, non-production data to keep the demonstration ethical and safe.

10. Project Timeline

Incremental delivery with a demonstrable workflow at the end of every phase. Verification is a real clinical run rather than a passing test suite.

Phase	Focus	Milestone / Deliverable	Duration
P0	Foundation and identity	A patient registers by phone, receives an OTP and authenticates; a forged identity header is rejected	3 wk
P1	The visit loop	A walk-in registered at reception reaches the patient's phone in two seconds and advances everywhere at once	9 wk
P2	Clinical workflow	The patient sheet reaches the doctor first; the consultation produces a signed prescription	6 wk
P3	Commerce and laboratory	Consultation to invoice to payment to dispensing, with stock decremented exactly once	6 wk
P4	AI platform	The AI sheet matches or beats the deterministic baseline on allergy and medication recall	5 wk
P5	Communications, analytics	Every queue change reaches the correct channel per patient preference	2 wk
P6	Hardening	Penetration test closed, 500 concurrent watchers sustained, recovery inside one hour	3 wk
Total, with the deployable release at the end of P3			34 wk

11. References

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12. Group Member Details

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